Hes·so VALAIS WALLIS

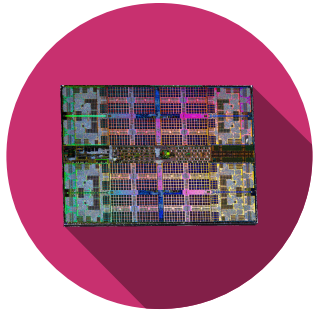
School of Engineering

Computer Architecture

Introduction

Ind

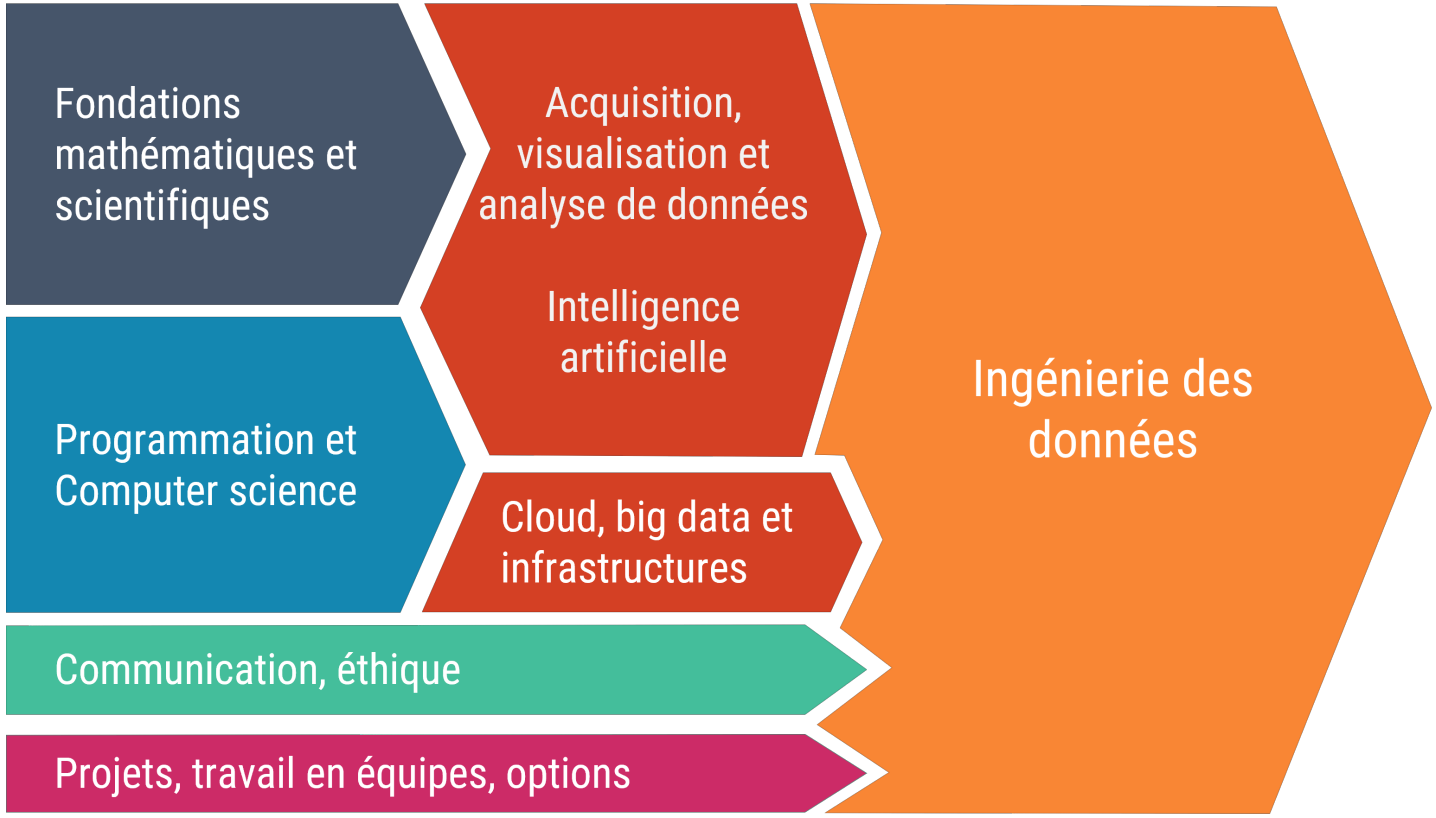
Information and Communication Systems program



Silvan Zahno silvan.zahno@hevs.ch

Modules

ISC



BSc en Informatique et Systèmes de communication

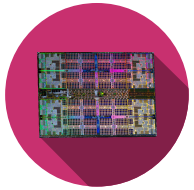
Modules

First Year

1ère année / 1. Jahr

Cours *Vorlesungen*	*Heures* *Stunden*	*Crédits* *Credits*
Semestre 1 / Semester 1	**32**	**28**
Algèbre linéaire 1 / Lineare Algebra 1	4	4
Analyse 1 / Analysis 1	6	5
Communication 1 / Kommunikation 1	2	2
Ethique et aspects juridiques / Ethik und Rechtaspekte	2	2
Langues 1 / Sprachen 1	4	3
Programmation impérative / Imperative Programmierung	8	7
Systèmes numériques / Digitale Technik	*6*	*5*
Semestre 2 / Semester 2	**36**	**32**
Algèbre linéaire 2 / Lineare Algebra 1	4	4
Analyse 2 / Analysis 2	6	5
Architecture des ordinateurs / Computerarchitektur	4	*3*
Communication 2 / Kommunikation 2	2	2
Intro. à la sécurité / Sicherheitbewusstsein	4	3
Langues 2 / Sprachen 2	4	3
Programmation objet / Objektorienterte Programmierung	8	6
Réseaux / Netzwerke	4	6
Projet 1 / Projekt 1	Summer school	3

Goal of the course



Understand the basic structure of modern computers and its connection with programming.

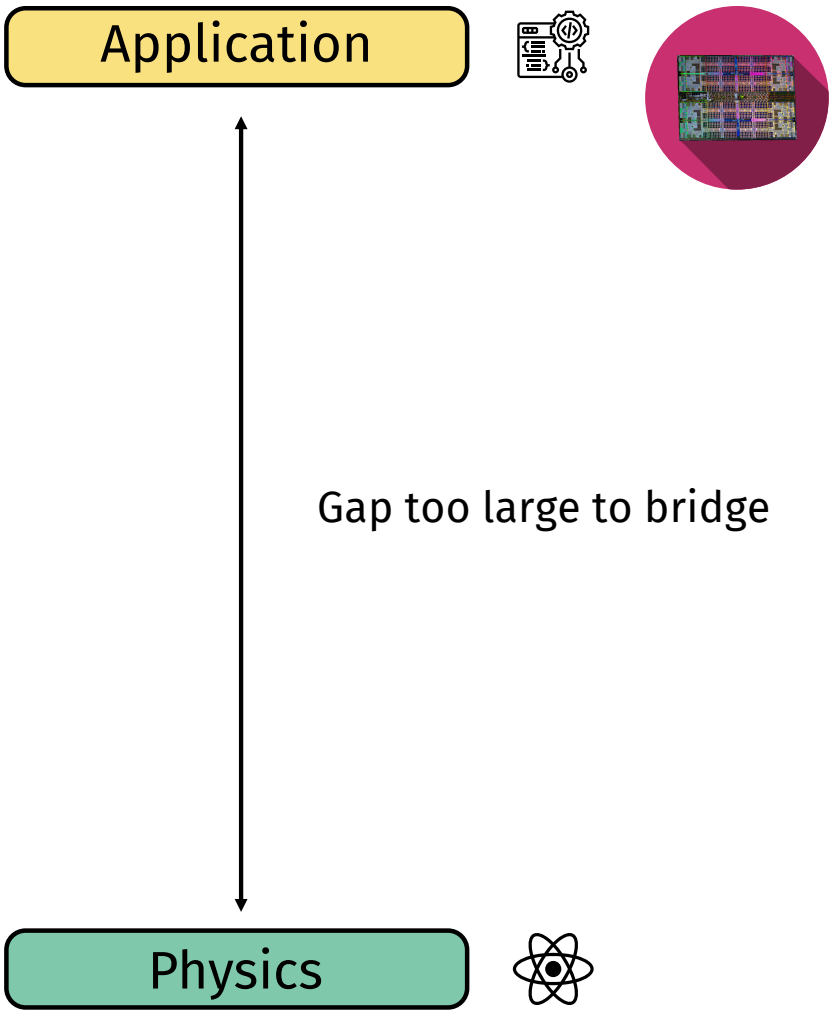
This means:

- Methodical basics: beginning of the semester
- Simple system: end of the semester

What is Computer Architecture?

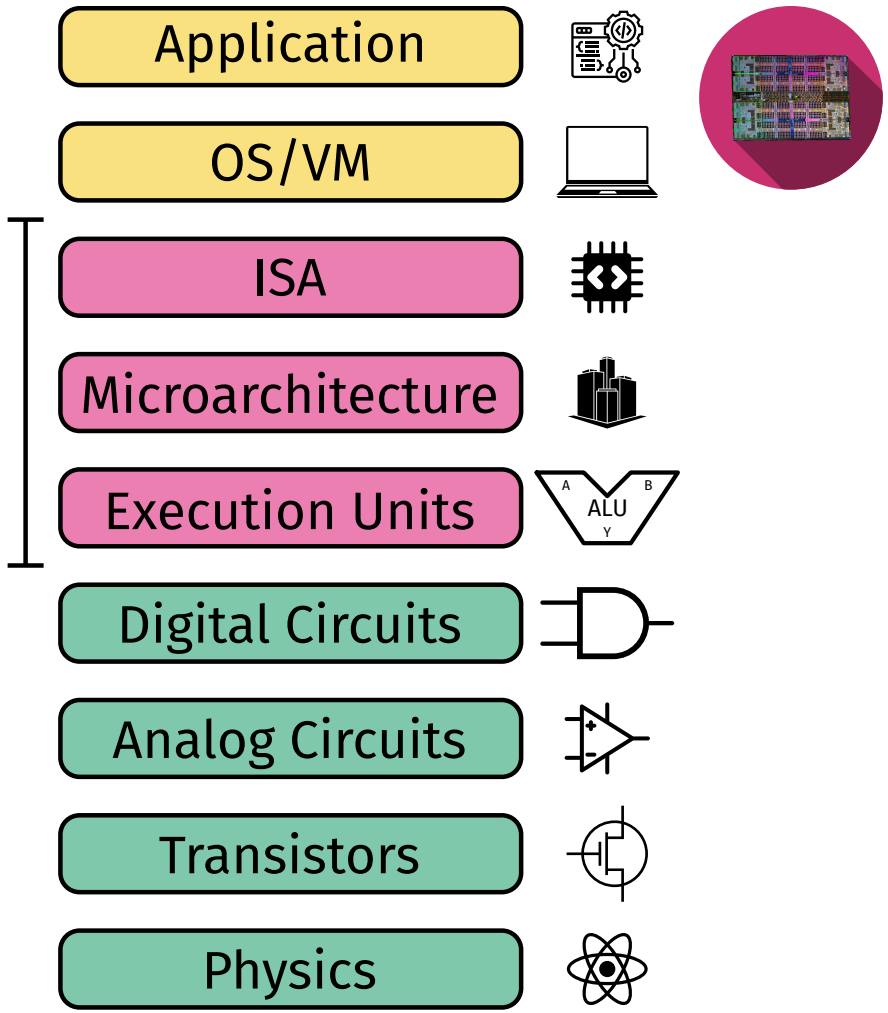
The gap

In the most general definition, computer architecture is the design of the abstraction and implementation layers that allow us to execute information processing applications efficiently using manufacturing technologies



What is Computer Architecture?

The different layers



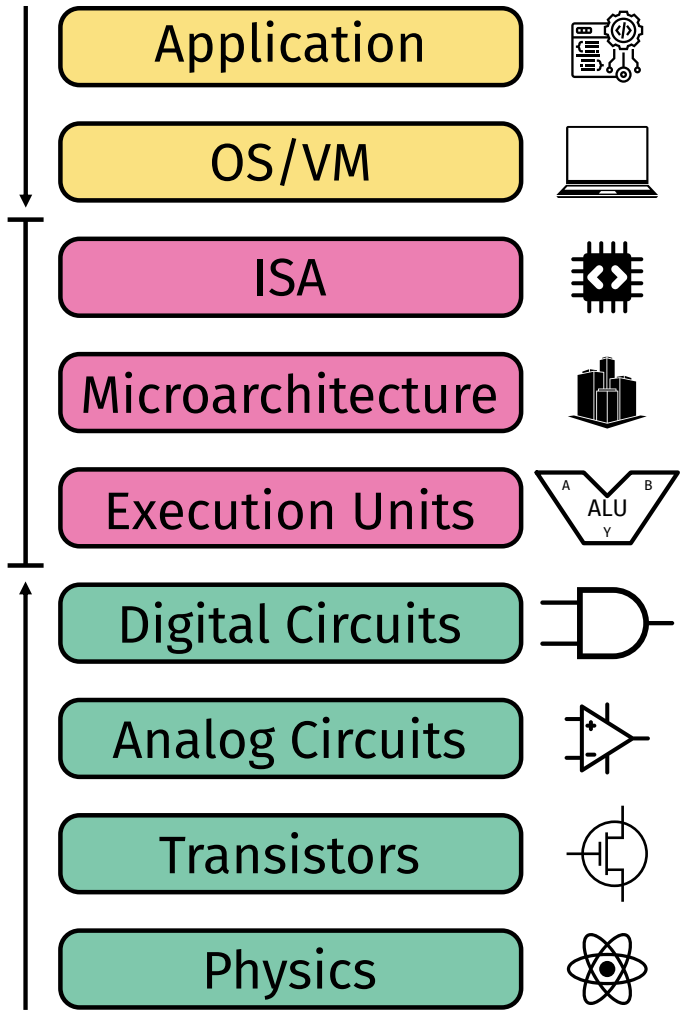
What is Computer Architecture?

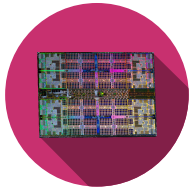
Constantly changing

Application Requirements

Car
We are here

Technology Constraints





Computer Architecture in practice

Ariane 5

[1]

https://youtu.be/PK_yguLapgA

Application Areas

Organisation

Contact points

Professors

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Tel: +41 58 606 88 07

Collaborators

Amand Axel (AmA)
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Email: axel.amand@hevs.ch
Tel: +41 58 606 87 43

Organisation

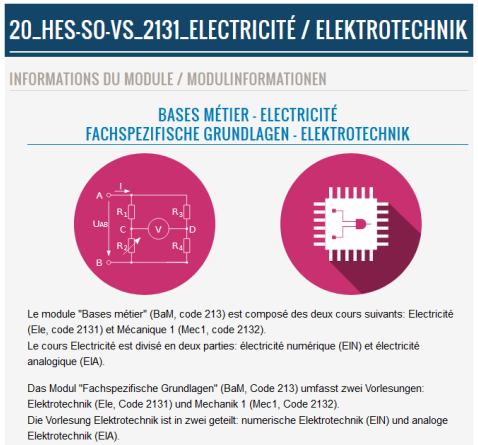
- Cours - 2h/week
- Labor - 2h/week
- Project - last 5-6 weeks
- Exam
  - Labo evaluation
  - Project evaluation
  - Exam at the end of the semester (only summary of 2 pages allowed)

	Fallsemester (EIN)			*Springsemester (CAr)*		
Evaluation	Exa 1	Project	Exa Sem	Labo	Project	Exa
Coefficient	0.5	0.5	1	0.5	0.5	1
Semestergrade	1		1	1		1
Modulegrade	5/9			4/9		

Documents and Informations

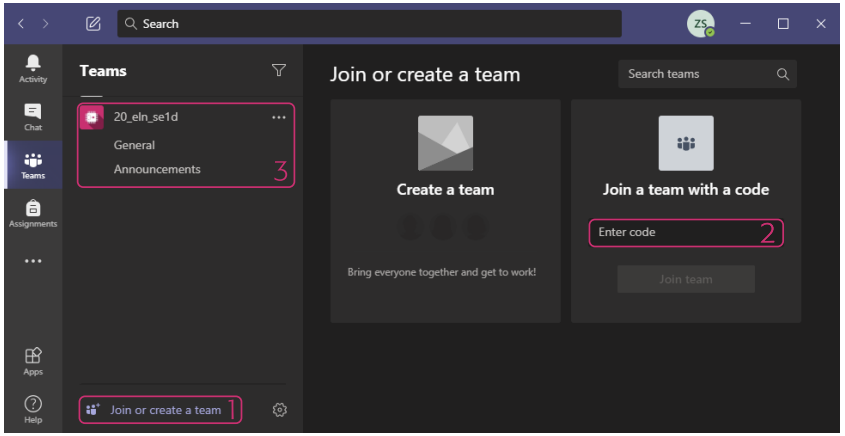
Moodle Cyberlearn

- https://isc.hevs.ch/learn/course/view.php?id=12
- 102.2 – ArchOrd
- Password: `welcome`

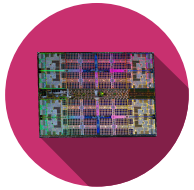


Microsoft Teams

- ISC-CAr
- Access Code: `tah5o6p`
- Channels
  - Announcements – Message of the module management
  - General – Class chat

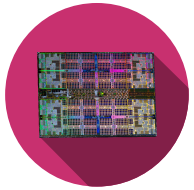


What programming languages do you know?



[2]

What programming languages do you know?



High-level languages:

Python

Ruby

JavaScript

PHP

Swift

Java

C#

Objective-C

TypeScript

Kotlin

Dart

Go

Low-level languages:

Rust

C

C++

Fortran

Ada

COBOL

Lisp

Assembly languages

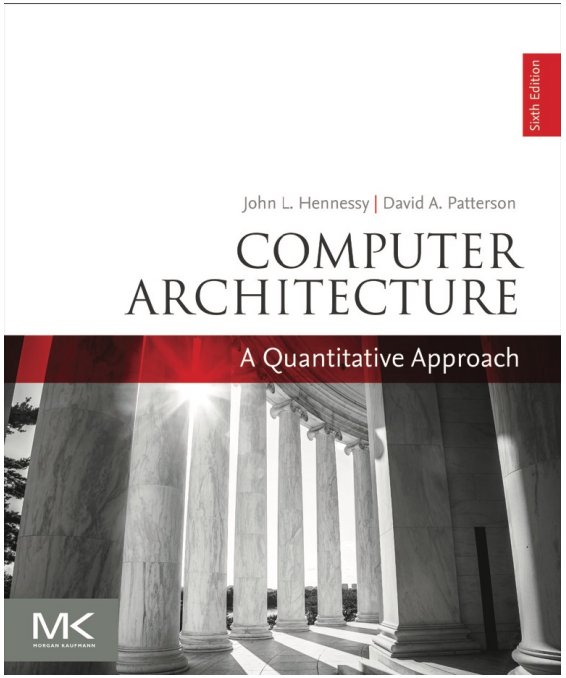
- x86
- ARM
- MIPS
- RISC-V

Additional Books

Computer Architecture – A Quantitative Approach

John L. Hennessy & David A. Patterson

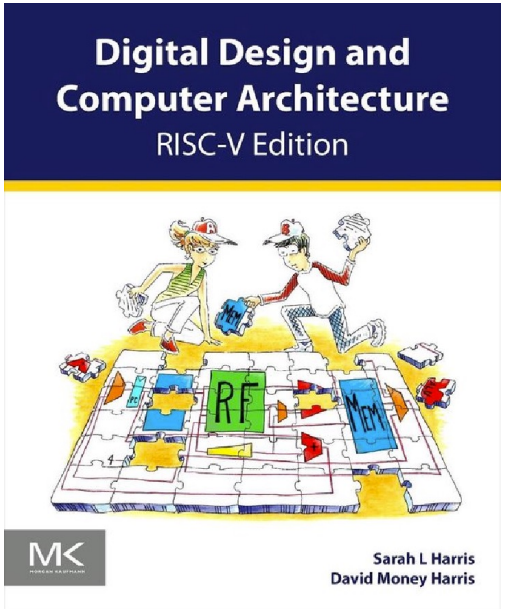
ISBN: 978-0-12-811905-1



Digital Design and Computer Architecture

Sarah L. Harris & Davod Money Harris

ISBN: 978-0-12-820064-3



References

[1]

"Ariane 5 rocket launch explosion - YouTube." https://www.youtube.com/watch?v=PK_yguLapgA&t=142s (accessed Jul. 18, 2022).

[2]

J. L. Hennessy and D. A. Patterson, *Computer Architecture: A Quantitative Approach*, 6th Edition. Elsevier, 2019.

[3]

HEVs, "Formation Bachelor Ingénieur·e en I... | HES-SO Valais-Wallis," *La HES-SO Valais-Wallis propose 10 filières de formation dans 5 Hautes Ecoles: Ingénierie, Gestion, Santé, Travail Social et Art*. https://www.hevs.ch/fr/hautes-ecoles/haute-ecole-d-ingenierie/informatique-et-systemes-de-communication/formation-bachelor-ingenieure-en-informatique-et-systemes-de-communication-201140 (accessed Jul. 18, 2022).

[4]

Prof. Dr. Ben H. Juurlink, *TU Berlin: Course AES*, (Oct. 22, 2018). Accessed: Jul. 18, 2022. [Online Video]. Available: https://www.youtube.com/channel/UCPSsA8oxlSBjidJsSPdpjsQ

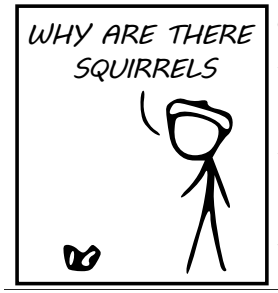
QUESTIONS

CAN BE ASKED BY ANYONE ANYTIME

WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SVCHOST.EXE RUNNING
WHY AREN'T ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY WUBA LUBBA DUB DUB MEANING
WHY IS THERE A WHALE AND A POT FALLING
WHY ARE THERE SO MANY BIRDS IN SWISS
WHY IS THERE SO LITTLE RAIN IN WALLIS
WHY IS WALLIS WEATHER FORECAST ALWAYS WRONG
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
HOW FAST IS LIGHTSPEED
WHY ARE OLD KLINGONS DIFFERENT

WHY AREN'T THERE DINOSAUR GHOSTS

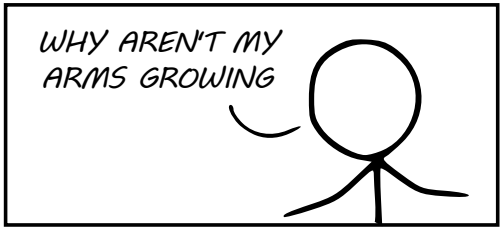
WHY IS THERE HELL IF G



WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY HAVE DINOSAURS NO FUR
WHY DO IGUANAS DIE

WHY ARE SWISS AFRAID OF DRAGONS

WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON TWITTER
WHY IS HTTPS IMPORTANT



WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY

WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY IS 42 THE ANSWER TO EVERYTHING
WHY CAN'T NOBODY ELSE LIFT THORS HAMMER
WHY IS MARVIN ALWAYS SO SAD
WHY IS THERE LAVA
WHY IS THERE A SWARM OF ANTS
WHY IS THERE PILEGRIM
WHY ARE THERE SO MANY CROWS IN ROCHESTER
WHY IS TO BE OR NOT TO BE FUNNY
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE DOGS AFRAID OF FIRE
WHY IS THERE NO KING IN EN

WHY ARE THERE ANTS IN MY LAPTOP

WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN

WHY ARE THERE FEMALE

WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE FPGA's EVERYWHERE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOCKS
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO PEOPLE DIE

WHY IS LIFE SO

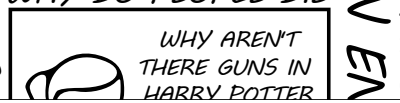
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPYDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY ARE THERE 5 GRADES

WHY

WHY 2*B | $\overline{2*B}$

WHAT IS https://xkcd.com/1256/

WHY DO THEY SAY T-MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMMING MORE ACIDIC



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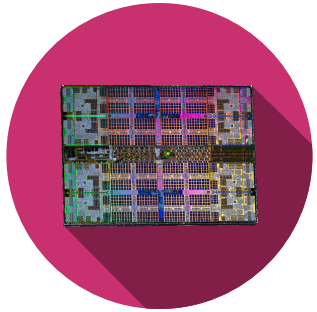

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